Smart Inventory Management System for Construction Sites

EXERCISE NO – 06

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6. Smart Inventory Management System for Construction Sites

Project Overview:

Construction sites use a large number of materials every day, including cement, steel rods, bricks, sand, tools, electrical components, pipes, and other resources. Managing these materials manually can be difficult because materials are continuously moved between the storage area and different work locations.

Traditional inventory management often depends on:

- Paper records.
- Manual counting.
- Registers.
- Verbal communication.
- Periodic stock checking.

These methods can lead to incorrect records, material loss, delays, overstocking, and inefficient resource utilization.

To address these problems, this project proposes a Smart Inventory Management System for Construction Sites using an ESP8266 NodeMCU and RFID technology.

The system uses an MFRC522 RFID reader to identify RFID cards/tags assigned to different construction materials. When a tag is scanned, the ESP8266 identifies the material and updates its inventory status.

An OLED/LCD display provides information about the scanned material and its current inventory status. A buzzer provides an audible confirmation whenever a valid RFID card is detected.

Main Objective:

The main objective of this project is to develop a low-cost smart inventory system that can:

- Identify construction materials using RFID.
- Record material movement.
- Reduce manual inventory errors.
- Display material information in real time.
- Provide instant feedback when an RFID card is scanned.
- Improve tracking of construction-site resources.
- Reduce material loss and unauthorized movement.

<u>Problem Statement:</u>

Construction projects require effective management of materials because materials represent a significant portion of project resources and costs.

In a conventional construction site, materials are often recorded manually when they are received, issued, or returned.

<u>Manual Record-Keeping</u>

Workers may need to write down:

- Material name.
- Quantity.
- Date.
- Time.
- Person receiving the material.
- Material movement status.

Manual recording takes time and can result in missing or incorrect information.

<u>Material Loss</u>

Construction materials may be moved between storage areas and work locations without proper records.

This makes it difficult to determine:

- Where the material went.
- When it was removed.
- Who handled it.
- Whether it was returned.

Human Errors

Incorrect entries, duplicate entries, forgotten entries, and incorrect material counts can affect inventory accuracy.

Poor Resource Utilization

If the site does not know its current inventory accurately, it may:

- Purchase materials unnecessarily.
- Run out of required materials.
- Store excessive quantities.
- Delay construction activities.

Proposed Problem Definition

Therefore, a simple automated system is required to identify construction materials and record their movement using electronic identification rather than depending entirely on manual records.

The proposed RFID-based system provides a foundation for automated inventory management.

Proposed Solution:

The proposed system uses RFID technology to identify materials.

The ESP8266 Node MCU acts as the main controller.

The MFRC522 RFID reader detects RFID cards/tags.

Each RFID card is assigned to a specific construction material.

For example:

RFID UID 1 → Cement

RFID UID 2 → Steel Rods

RFID UID 3 → Bricks

When a card is placed near the RFID reader:

- The RFID reader detects the card.
- The reader obtains the UID.
- The ESP8266 receives the UID.
- The ESP8266 compares the UID with stored values.
- The corresponding material is identified.
- The OLED displays the material information.
- The inventory movement is updated.
- The buzzer provides confirmation.

System Architecture:

The system consists of four major sections:

Input Section

The MFRC522 RFID reader receives information from RFID cards.

Processing Section

The ESP8266 identifies the RFID UID and performs inventory operations.

Output Section

The OLED/LCD displays:

- Material name.
- Movement status.
- Quantity.
- Remaining stock.

The buzzer provides audio confirmation.

Inventory Section

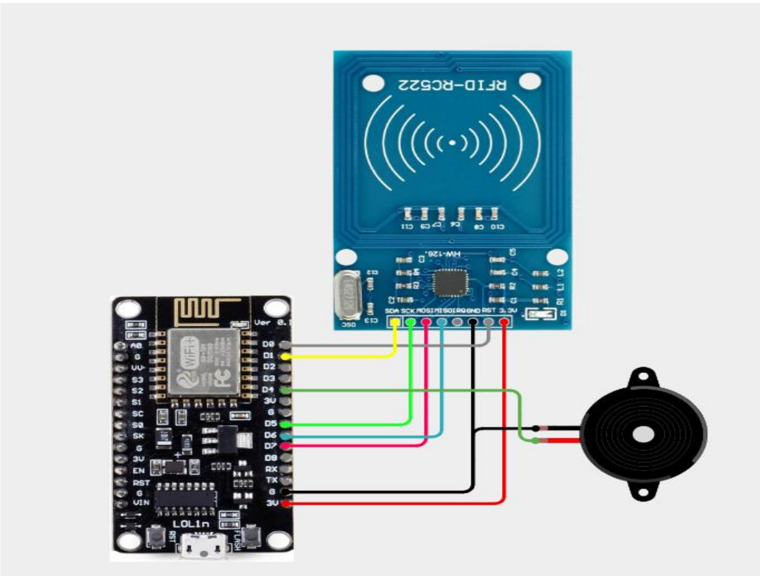
The ESP8266 maintains the current inventory values in its program memory during prototype operation.

For a more advanced version, the data can be transmitted through Wi-Fi to a database or cloud server.

<u>Circuit Table:</u>

OLED Pin	ESP8266 Pin	Function
VCC	3.3V	Power
GND	GND	Ground
SDA	D2 / GPIO4	I^2C Data
SCL	D1 / GPIO5	I^2C Clock

<u>Circuit Design:</u>



Algorithm:

- Start.
- Initialize ESP8266.
- Initialize SPI communication.
- Initialize MFRC522.
- Initialize OLED.
- Initialize buzzer.
- Load initial inventory.
- Display "Scan RFID Card".
- Wait for RFID card.
- Detect RFID card.
- Read RFID UID.
- Compare UID with registered RFID IDs.
- If UID is not recognized:
    - Display "Unknown Card".
    - Activate buzzer.
    - Return to scanning.
- Determine movement mode.
- Enter quantity.
- If movement is OUT:
    - Check available stock.
    - Reduce inventory quantity.
- If movement is IN:
    - Increase inventory quantity.
- Display updated inventory.
- Activate confirmation buzzer.
- Wait for next RFID card.
- Repeat continuously.

Coding:

```
#include <ESP8266WiFi.h>
#include <SPI.h>
#include <MFRC522.h>
#include "AdafruitIO_WiFi.h"



#define WIFI_SSID "K-APIL"
#define WIFI_PASS "kapil.2333"

#define IO_USERNAME "saravanakarthiksr"
#define IO_KEY      "aio_idPg82y91fmNVXL0iJA71zKwH97q"


AdafruitIO_WiFi io(IO_USERNAME, IO_KEY, WIFI_SSID, WIFI_PASS);

AdafruitIO_Feed *rfidUID =
  io.feed("rfid-uid");

AdafruitIO_Feed *materialName =
  io.feed("material-name");

AdafruitIO_Feed *inventoryStatus =
  io.feed("inventory-status");

AdafruitIO_Feed *rfidDetection =
  io.feed("rfid-detection");

AdafruitIO_Feed *scanCount =
  io.feed("scan-count");



#define SS_PIN     4      // GPIO4 = D2
#define RST_PIN    16     // GPIO16 = D0
#define BUZZER_PIN 2      // GPIO2 = D4

MFRC522 rfid(SS_PIN, RST_PIN);



int totalScans = 0;

// Example inventory status
bool cementStatus = true;
```

```
bool steelStatus = true;
bool bricksStatus = true;



void setup()
{
  Serial.begin(115200);

  pinMode(BUZZER_PIN, OUTPUT);
  digitalWrite(BUZZER_PIN, LOW);

  // Start SPI
  SPI.begin();

  // Start RFID
  rfid.PCD_Init();

  delay(500);

  Serial.println();
  Serial.println("=======================================");
  Serial.println(" SMART INVENTORY MANAGEMENT SYSTEM");
  Serial.println("=======================================");

  // Connect to Adafruit IO
  Serial.println("Connecting to Adafruit IO...");

  io.connect();

  while (io.status() < AIO_CONNECTED)
  {
    Serial.print(".");
    delay(500);
  }

  Serial.println();
  Serial.println("Adafruit IO Connected!");

  // Initial cloud status
  rfidDetection->save("NO TAG");
  inventoryStatus->save("READY");

  Serial.println("RFID Reader Ready");
  Serial.println("Scan RFID Tag...");
  Serial.println();
```

```
}

void loop()
{
  // Keep Adafruit IO connection alive
  io.run();

  // Check RFID card
  if (!rfid.PICC_IsNewCardPresent())
  {
    return;
  }

  // Read RFID card
  if (!rfid.PICC_ReadCardSerial())
  {
    return;
  }

  Serial.println("------------------------------------");
  Serial.println("RFID CARD DETECTED!");



  String uidString = "";

  for (byte i = 0; i < rfid.uid.size; i++)
  {
    if (rfid.uid.uidByte[i] < 0x10)
    {
      uidString += "0";
    }

    if (i > 0)
    {
      uidString += " ";
    }

    if (rfid.uid.uidByte[i] < 0x10)
    {
      uidString += "0";
    }

    uidString += String(rfid.uid.uidByte[i], HEX);
  }
```

```
uidString.toUpperCase();

Serial.print("RFID UID: ");
Serial.println(uidString);

 rfidDetection->save("DETECTED");

// Send UID to Adafruit IO
rfidUID->save(uidString);

// Increase scan count
totalScans++;
scanCount->save(totalScans);


String material = "UNKNOWN";
String status = "UNKNOWN";

if (uidString == "23 4A 91 7C")
{
  material = "CEMENT";

  if (cementStatus == true)
  {
    status = "OUT";
    cementStatus = false;
  }
  else
  {
    status = "IN";
    cementStatus = true;
  }
}

else if (uidString == "11 22 33 44")
{
  material = "STEEL";

  if (steelStatus == true)
  {
    status = "OUT";
    steelStatus = false;
  }
  else
  {
    status = "IN";
```

```
    steelStatus = true;
  }
}

else if (uidString == "55 66 77 88")
{
  material = "BRICKS";

  if (bricksStatus == true)
  {
    status = "OUT";
    bricksStatus = false;
  }
  else
  {
    status = "IN";
    bricksStatus = true;
  }
}

else
{
  material = "UNKNOWN";
  status = "UNAUTHORIZED";
}


Serial.print("Material: ");
Serial.println(material);

Serial.print("Inventory Status: ");
Serial.println(status);


materialName->save(material);
inventoryStatus->save(status);



if (material == "UNKNOWN")
{
  // Two beeps for unknown RFID
  beepBuzzer(2);
}
else
{
```

```
    // One beep for valid RFID
    beepBuzzer(1);
  }

  Serial.println("Data sent to Adafruit IO");
  Serial.println("------------------------------------");

  // Stop RFID communication
  rfid.PICC_HaltA();
  rfid.PCD_StopCrypto1();

  delay(1000);

  // Reset cloud detection status
  rfidDetection->save("NO TAG");
}

void beepBuzzer(int times)
{
  for (int i = 0; i < times; i++)
  {
    digitalWrite(BUZZER_PIN, HIGH);
    delay(200);

    digitalWrite(BUZZER_PIN, LOW);
    delay(200);
  }
}
```

System Working:

When power is supplied, the ESP8266 initializes the RFID reader, OLED, SPI communication, and buzzer.

The OLED displays:

SMART INVENTORY

RFID READY

Scan Card...

RFID Detection:

A worker brings an RFID card close to the MFRC522 reader.

The reader detects the card and obtains its UID.

For example:

UID = 93 7A 2C 19

The ESP8266 compares this UID with the registered UID list.

Material Identification:

If the UID matches the Cement card:

RFID UID

↓

Cement

The ESP8266 identifies the material as Cement.

The OLED displays:

SMART INVENTORY

Material: CEMENT

Movement: OUT

Stock: 50

Material OUT Operation:

Suppose one unit of Cement is issued.

Before transaction:

Cement Stock = 50

After RFID scan:

Cement Stock = 49

The OLED displays:

CEMENT

OUT SUCCESS

Stock Updated

The buzzer produces a short confirmation beep.

Material IN Operation:

When materials are received or returned, the system can operate in IN mode.

Suppose the Cement stock is:

49 bags

and one bag is returned.

The system updates:

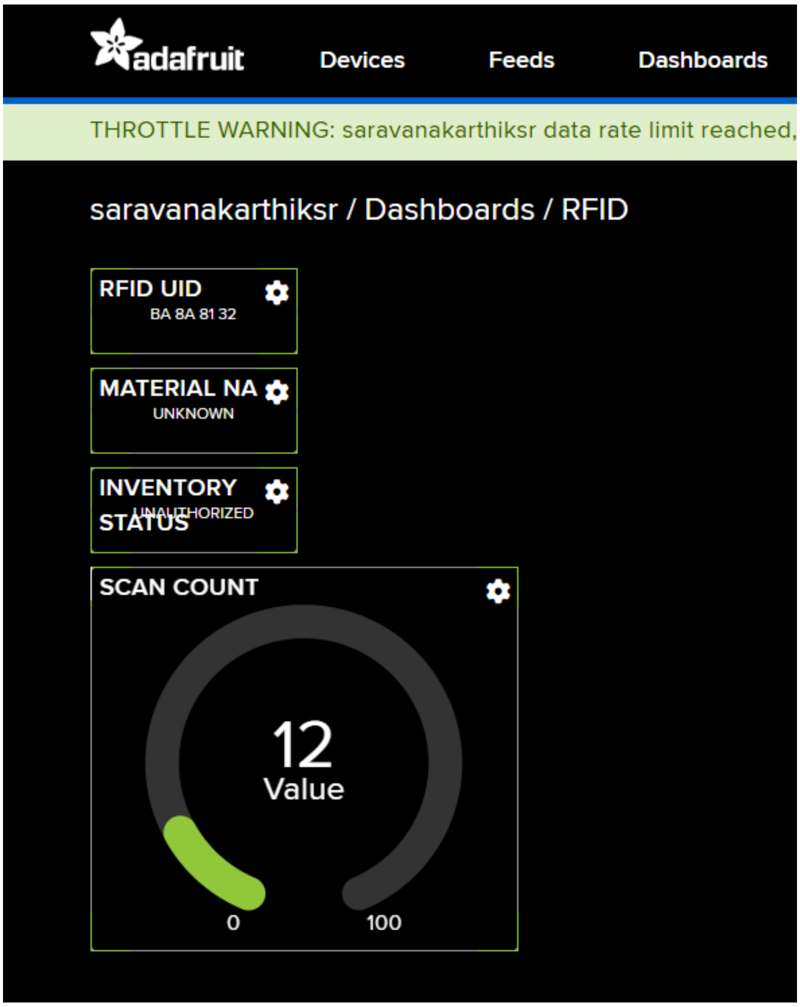
49 + 1 = 50 bags

The OLED displays:

CEMENT

IN SUCCESS

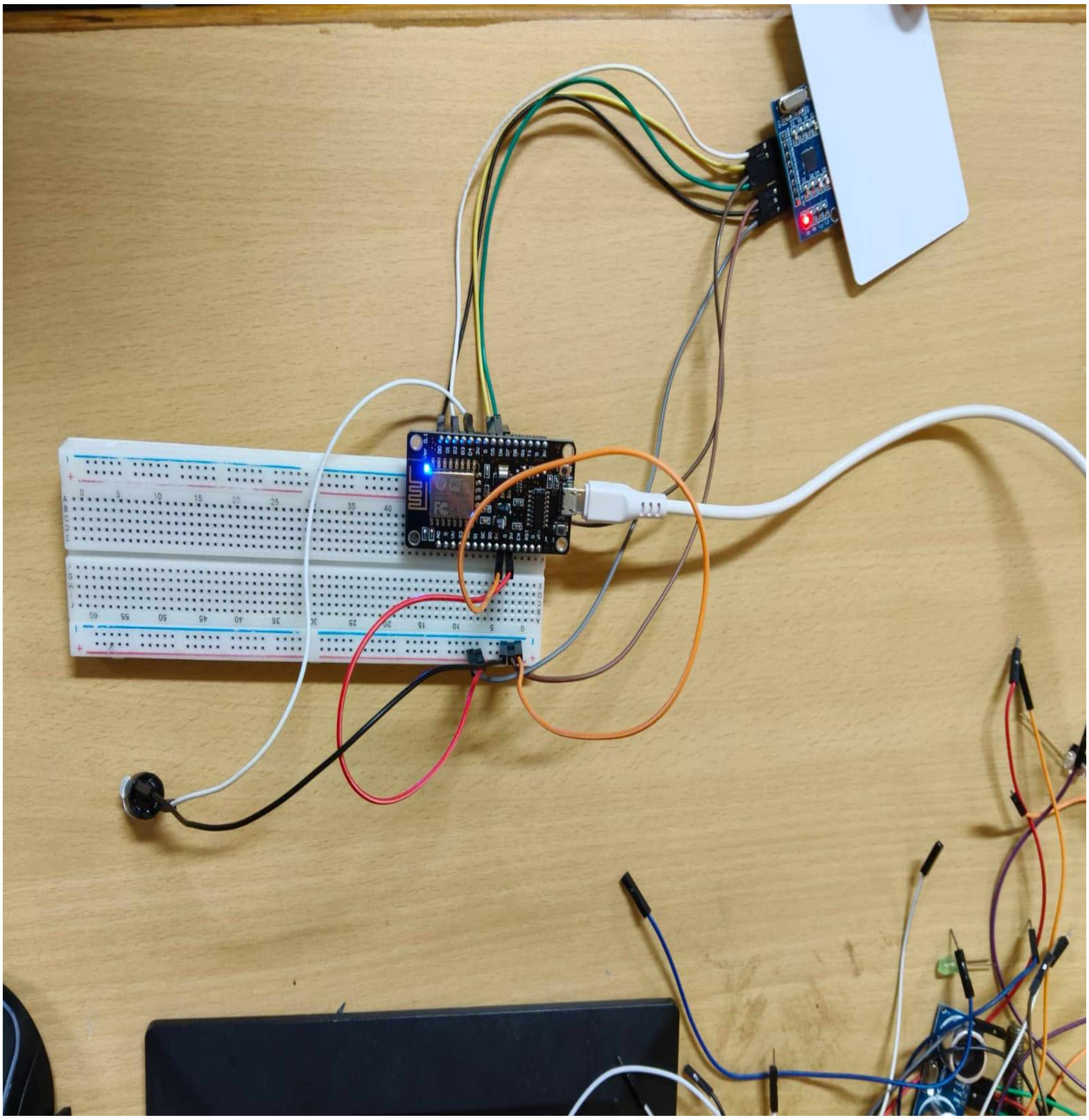
Stock Updated



Bill of Materials:

Component	Quantity
ESP8266 NodeMCU	1
RFID Reader (MFRC522)	1
RFID Tags/Cards	3
OLED/LCD Display	1
Buzzer	1
Breadboard	1
Jumper Wires	1 Set

Prototype Implementation:



Results & Performance Analysis :

RFID Identification

The RFID reader provides fast electronic identification without requiring the worker to manually enter a material name.

Inventory Accuracy

Automated updates reduce the possibility of manual calculation errors.

Response Time

The system can identify a compatible RFID card and display its information within a short period after scanning.

User Feedback

The OLED provides visual feedback while the buzzer provides audible confirmation.

Unauthorized Card Detection

The system can distinguish registered RFID cards from unregistered cards.

Resource Utilization

By maintaining inventory information, the system can help construction-site managers understand which materials are available and which materials are being consumed.